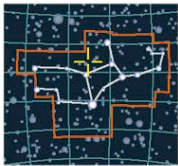


PULSATING VARIABLE

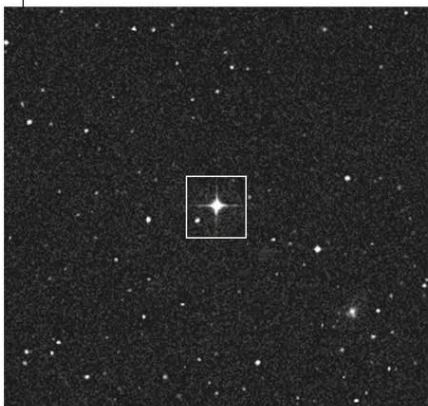
W Virginis



DISTANCE FROM SUN
10,000 light-years
MAGNITUDE 9.6
SPECTRAL TYPE F0

VIRGO

W Virginis lends its name to a class of variable stars that are similar to Cepheid variables (see p.282) and are also known as Population II Cepheids. W Virginis is a pulsating yellow giant star. The outer layers of its atmosphere expand and contract with a period of 17.27 days. The period has lengthened over the last 100 years of observation. The pulsation causes a 1.2-magnitude variation, as the star doubles its size during the cycle. As a Population II star (see p.227), W Virginis is among the oldest stars in the Milky Way.

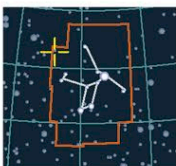


W VIRGINIS

W Virginis is located high above the galactic plane in the diffuse halo of old stars that surrounds the Milky Way (see pp.226–229). Like other W Virginis variables, it is an old Population II star, on average lower in mass and magnitude than a Cepheid variable.

PULSATING VARIABLE

RR Lyrae



DISTANCE FROM SUN
744 light-years
MAGNITUDE 7.1
SPECTRAL TYPE F5

LYRA

RR Lyrae is the brightest member of the class of variables that takes its name. RR Lyrae stars are similar to Cepheid variables (see p.282) but are less luminous and tend to have shorter periods—ranging from about 5 hours to just over a day. RR Lyrae's period is 0.567 days, and its magnitude varies between 7.06 and 8.12. By comparing the luminosity of RR Lyrae variables with their apparent magnitude, a good distance determination can be made. In this way, RR Lyrae variable stars are important tools for calculating astronomical distances.

BRIGHT VARIABLE

RR Lyrae has an average luminosity 40 times that of the Sun and a surface temperature of about 12,000°F (6,700°C). RR Lyrae stars are often found in globular clusters, and they are sometimes referred to as cluster variables.



PULSATING VARIABLE

Delta Cephei



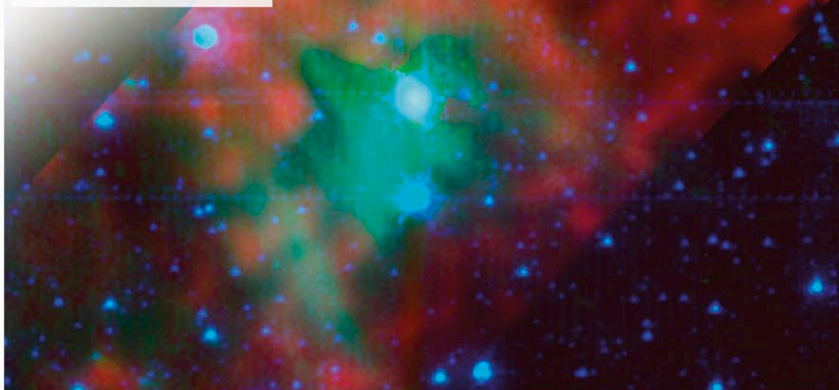
DISTANCE FROM SUN
982 light-years
MAGNITUDE 4
SPECTRAL TYPE F5

CEPHEUS

Delta (δ) Cephei is the prototype of the Cepheid class of variable stars (see p.282), and to astronomers it is one of the most famous stars in the sky. Its magnitude variation, from 3.48 to 4.37, is visible to the naked eye, and its short period of 5 days, 8 hours, and 37.5 minutes makes it a popular target for amateur observers. Its position in the sky makes it easy to find, and it is close to two comparison stars with magnitudes at the ends of Delta Cephei's range. Delta Cephei is a supergiant with a spectral type that varies between F5 and G2.

BOW SHOCK

Strong stellar winds blowing out from Delta Cephei excite interstellar gas and dust in this infrared image.



EXPLORING SPACE

THE CEPHEID PROTOTYPE

In 1921, Henrietta Leavitt (1868–1921), an astronomer based at the Harvard Observatory, discovered a strong link between the period and luminosity of a group of stars later known as Cepheid variables (see p.282), of which Delta Cephei was the prototype. This correlation provided astronomers with a new way of measuring distances in space. In 1923, Edwin Hubble used it to prove that the Andromeda Galaxy is

situated outside the Milky Way. Since then, Cepheids have provided more useful information about the universe than any other star type.



HENRIETTA LEAVITT

PULSATING VARIABLE

Zeta Geminorum



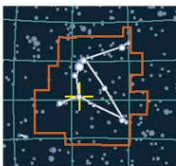
DISTANCE FROM SUN
1,168 light-years
MAGNITUDE 4
SPECTRAL TYPE G0

GEMINI

Also known as Mekbuda, Zeta (ζ) Geminorum is a yellow supergiant about 3,000 times as luminous as the Sun. It is one of the easiest Cepheid variable stars to observe (see p.282). Zeta Geminorum, like all Cepheids, is unstable and pulsates, changing its temperature, size, and spectral type. It has a period of 10.2 days and a magnitude that varies from 3.6 to 4.2. Its period is shortening at the rate of about 3 seconds per year. Zeta Geminorum is also a binary star with a faint, magnitude-10.5 companion.

PULSATING VARIABLE

Eta Aquilae



DISTANCE FROM SUN
1,400 light-years
MAGNITUDE 3.9
SPECTRAL TYPE F6

AQUILA

Eta (η) Aquilae is a yellow supergiant star with a luminosity 3,000 times that of the Sun. It is one of the brightest Cepheid variables in the night sky (see p.282), and also one of the first to be discovered. Eta Aquilae varies in magnitude from 3.5 to 4.4 over a period of 7.2 days. Such a brightness variation is easily detectable with the naked eye. The magnitude range, by coincidence, is the same as the prototype of the class, Delta Cephei (above). Over this period, Eta Aquilae also varies in spectral type between G2 and F6.

NOVA

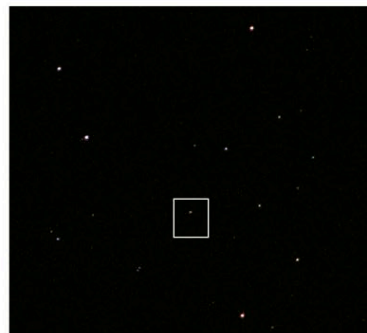
T Coronae Borealis



DISTANCE FROM SUN
2,025 light-years
MAGNITUDE 11
SPECTRAL TYPE M3

CORONA BOREALIS

T Coronae Borealis (T CrB), also known as the Blaze Star, is a recurrent nova (see p.282). It has displayed two major outbursts, one witnessed in 1866 and the other in 1946. T CrB's usual apparent magnitude is 10.8, but during outbursts, it has reached 2nd or 3rd magnitude. T CrB is a spectroscopic binary consisting of a red giant of spectral type M3 and a smaller blue-white dwarf. T CrB is usually about 50 times as luminous as the Sun, but during outbursts, it becomes more than 200,000 times as luminous. In between the outbursts,



BLAZE STAR

Although T Coronae Borealis cannot usually be seen without a telescope, during eruptions, it has "blazed" bright enough to be seen with the naked eye.

stellar dust and gas from the outer layers of the red giant are drawn onto the white dwarf. Eventually, the total mass of the white dwarf reaches a critical level, causing the outer layers of the white dwarf to explode violently. After the explosion, the two stars return to normal, to repeat the process many years later.